



BAULKHAM HILLS HIGH SCHOOL

Half -Yearly 2013
YEAR 12 TASK 2

Mathematics Extension 1

General Instructions

- Reading time – 5 minutes
- Working time – 2 hours
- Write using black or blue pen
- Board-approved calculators may be used
- Show all necessary working in Questions 11-14
- Marks may be deducted for careless or badly arranged work

Total marks – 70

Exam consists of 7 pages.

This paper consists of TWO sections.

Section 1 – Pages 2&3 (10 marks)

Questions 1-10

- Attempt Questions 1-10

Section II – Pages 4-7 (60 marks)

- Attempt questions 11-14

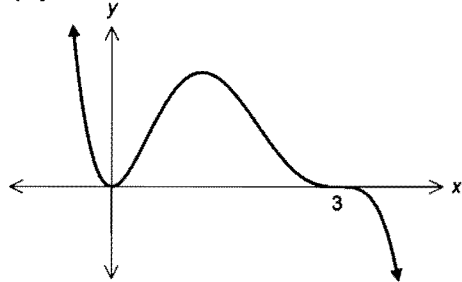
Table of Standard Integrals is on page 8

Section I - 10 marks

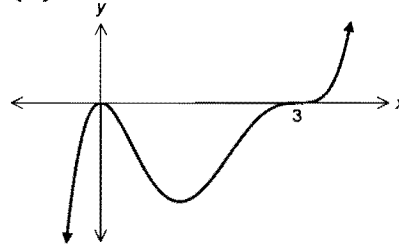
Use the multiple choice answer sheet for question 1-10

1. The graph of $y = x^2(3 - x)^3$ is best represented by:-

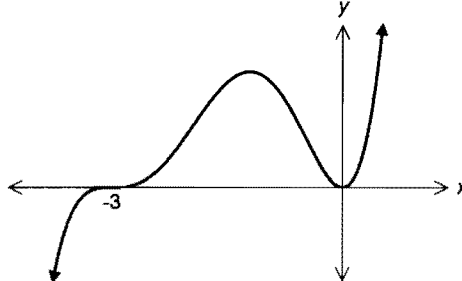
(A)



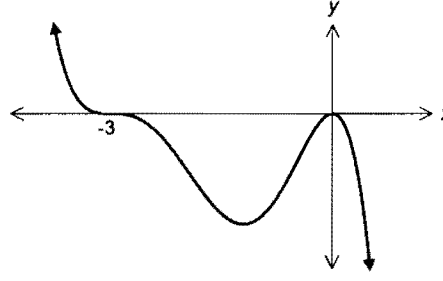
(B)



(C)



(D)



2. The acute angle to the nearest degree between the lines $3x + 2y = 6$ and $x - 4y + 2 = 0$ is:-

(A) 20°

(B) 42°

(C) 48°

(D) 70°

3. The solution to the inequality

$$\frac{x}{2+3x} \leq 1 \text{ is :-}$$

(A) $x \leq -1$ or $x \geq \frac{-2}{3}$

(B) $x \leq \frac{-3}{2}$ or $x \geq -1$

(C) $x \leq -1$ or $x > \frac{-2}{3}$

(D) $\frac{-2}{3} < x \leq \frac{-1}{2}$

4. The equation of the normal to the curve $y = (e^x + x)^3$ at the point where $x = 0$ is:-

(A) $x + 6y = 0$

(B) $x + 6y - 6 = 0$

(C) $6x - y + 1 = 0$

(D) $x + 3y - 3 = 0$

5. The point $P(x, y)$ is on the interval AB such that $3AP = 4PB$. If A has co-ordinates $(-1, 2)$ and B has co-ordinates $(6, -5)$ what are the co-ordinates of P ?

(A) $(2, -1)$

(B) $(3, -2)$

(C) $(-22, 23)$

(D) $(-27, 26)$

6. How many different arrangements of all of the letters of the word *POLYNOMIAL* are there if the *L*'s are separated?

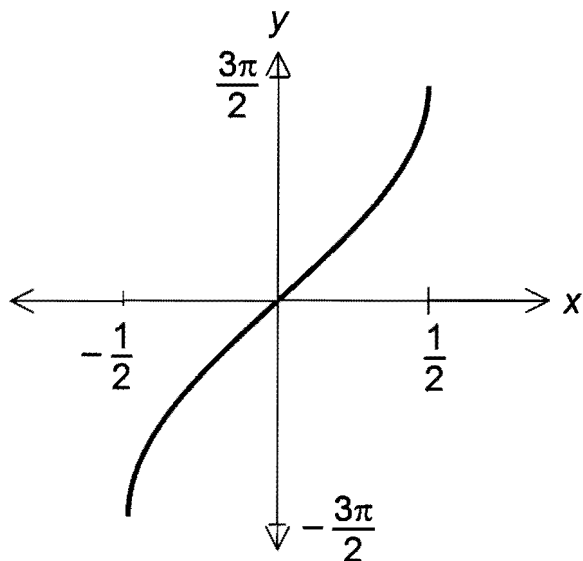
(A) 907200

(B) 181440

(C) 725760

(D) 1451520

7.



This inverse trigonometric graph has which of the following equations ?

(A) $y = \frac{1}{3} \sin^{-1} \left(\frac{x}{2} \right)$

(B) $y = 3 \sin^{-1} \left(\frac{x}{2} \right)$

(C) $y = \frac{1}{3} \sin^{-1}(2x)$

(D) $y = 3 \sin^{-1}(2x)$

8.

$$\int \frac{dx}{9 + 4x^2}$$

(A) $\frac{1}{8} \ln(9 + 4x^2) + c$

(B) $\frac{1}{4} \tan^{-1} \left(\frac{2x}{3} \right) + c$

(C) $\frac{1}{6} \tan^{-1} \left(\frac{2x}{3} \right) + c$

(D) $\frac{1}{9} \tan^{-1} \left(\frac{4x}{9} \right) + c$

9. The area between the curve $y = 2 + \cos x$, the x axis, $x = \frac{\pi}{2}$ and $x = \pi$ is rotated about the x axis. The volume of the solid generated is :-

(A) $\frac{9\pi - 16}{4}$

(B) $\frac{9\pi^2}{4}$

(C) $\frac{9\pi^2 + 16\pi}{4}$

(D) $\frac{9\pi^2 - 16\pi}{4}$

10. $\int 2^{2x+4} dx =$

(A) $\frac{2^{2x+4}}{2} + c$

(B) $\ln 2 \cdot 2^{2x+4} + c$

(C) $\frac{2^{2x+4}}{\ln 2} + c$

(D) $\frac{2^{2x+4}}{\ln 4} + c$

End of Section 1

Section II – Extended Response

Attempt questions 11-14.

Answer each question on the appropriate page in your exam booklet.

All necessary working should be shown in every question.

Question 11 (15 marks)

Marks

- a) If $P(x) = ax^3 - bx^2 + 6$ is a monic polynomial and when divided by $x - 4$ the remainder is -2 , find a and b .

3

- b) (i) Express $\sqrt{3} \cos x + \sin x$ in the form $A \cos(x - \alpha)$.

2

- (ii) Hence or otherwise solve $\sqrt{3} \cos x + \sin x = 1$ for $0^\circ \leq x \leq 360^\circ$.

2

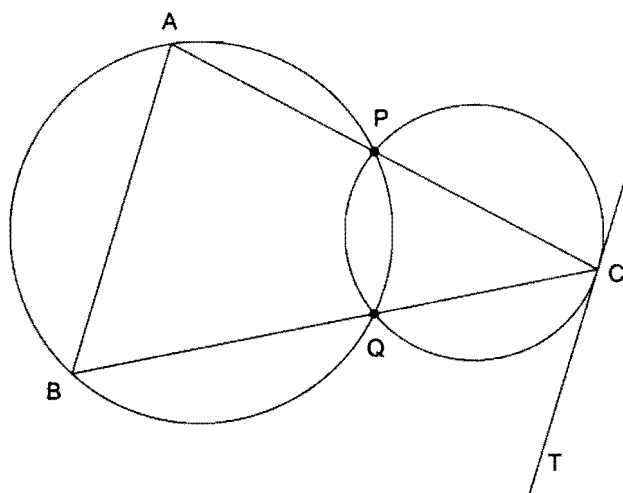
- c) Find $\int \cos x \sin^4 x \, dx$.

2

- d) If $\sin a = \frac{3}{4}$ and $90^\circ \leq a \leq 180^\circ$ and $\cos b = \frac{1}{5}$ and $0^\circ \leq b \leq 90^\circ$ find the exact value of $\sin(a + b)$.

3

- e)



LT is a tangent to the circle at C .

Prove that this tangent is parallel to AB .

3

Question 12 (15 marks)

- a) The cooling rate of a body is proportional to the difference between the temperature of a body (T) and the temperature of the surrounding medium (M).
ie. $\frac{dT}{dt} = k(T - M)$
- (i) Show that $T = M + Ae^{kt}$ is a solution to the above differential equation. 1
- (ii) A hot metal bar of 1000°C when immersed in cool water of 15°C for 5 minutes cools to 800°C . Find the temperature of the bar after a further 10 minutes. 2
- b) There are 4 teams of 8 players. The players of each team are numbered 1 to 8. A group of 6 players is to be selected from these 4 teams. 2
What is the probability that the group of 6 players chosen will have 4 players with the same number?
- c) (i) What is the largest domain containing $x = 4$ for $y = x^2 - 4x$ to have an inverse function? 1
- (ii) Sketch $y = x^2 - 4x$ for this domain and its inverse function on the same set of axes. 2
- (iii) What is the domain of the inverse function $y = f^{-1}(x)$? 1
- (iv) Find the equation of the inverse function as a function of x . 3
- (v) Find the point of intersection of the two curves. 2
- (vi) Evaluate $f^{-1}(f(1))$. 1

Question 13 (15 marks)

Marks

a) Prove by mathematical induction that $7^n + 3^n$ is a multiple of 10 if n is an odd positive integer.

3

b) If α, β, γ are the roots of $x^3 - 5x^2 + 4x - 2 = 0$, find the values of :-

(i) $\alpha + \beta + \gamma$.

1

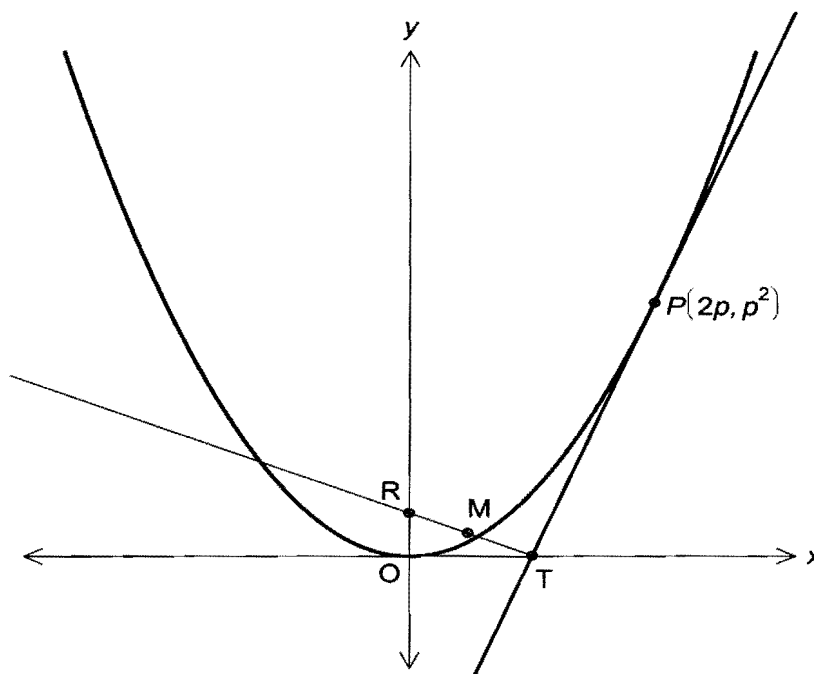
(ii) $\alpha\beta + \alpha\gamma + \beta\gamma$.

1

(iii) $\alpha^2 + \beta^2 + \gamma^2$.

1

c)



1

2

(i) Show the Cartesian equation for the parabola above is $x^2 = 4y$.

(ii) Show that the equation of the tangent at P is given by:
 $y = px - p^2$.

(iii) The tangent cuts the x axis at T and a perpendicular to this tangent is drawn from T to cut the y axis at R .

(I) Show that R is the focus of the parabola.

3

(II) What is the locus of M , the midpoint of RT ?

1

(III) Find the area of the triangle TPR in terms of p .

2

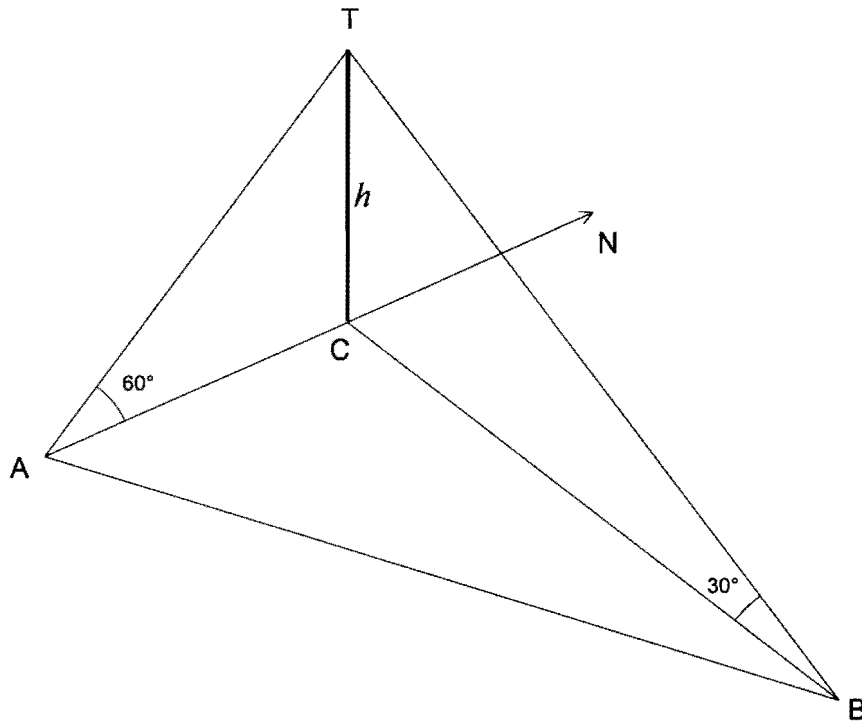
Question 14 (15 marks)

a) (i) Prove $\tan \theta = \sqrt{\frac{1-\cos 2\theta}{1+\cos 2\theta}}$ for $\tan \theta \geq 0$ 2

(ii) Hence find the exact value of $\tan \frac{\pi}{8}$ in simplified form. 3

b) Show that $f(x) = x^3 + 2x - 4$ has only 1 root. 2

c)



From a point A a person notes the elevation to the top of a tower CT due north is 60° . After walking to B on a bearing of 120° he notes the angle of elevation is then 30° .

(i) If the height of the tower is h , find AC and BC in terms of h in exact form. 1

(ii) Find the ratio of the distance AB to the height of the tower h . Give your answer in exact form. 3

d) Given $\log_b(xy^3) = m$ and $\log_b(x^3y^2) = p$. 4

Find $\log_b(\sqrt{xy})$ in terms of m and p .

End of Exam

Section 1. M. Choice

1. A 2. D 3. C 4. B 5. B
6. C 7. D 8. C 9. D 10. D

Section 2

11 a) $P(x) = ax^3 - bx^2 + 6$

monic $\Rightarrow a=1$ ①

$P(4) = -2 \therefore -2 = (4)^3 - b(4^2) + 6$
 $\therefore -2 = 64 - 16b + 6$
 $-72 = -16b$

$b = \frac{-72}{-16}$

$\therefore a=1 \quad b = \frac{9}{2}$ ①

b) $\sqrt{3} \cos x + \sin x = A \cos(x-\alpha)$

RHS $A \cos x \cos \alpha + A \sin x \sin \alpha$

$\therefore A \cos \alpha = \sqrt{3} \quad A \sin \alpha = 1$

$\therefore \tan \alpha = \frac{1}{\sqrt{3}} \Rightarrow \alpha = 30^\circ$ ①

$A = \sqrt{(\sqrt{3})^2 + (1)^2}$

$A = 2$ ①

$\therefore \sqrt{3} \cos x + \sin x = 2 \cos(x-30^\circ)$

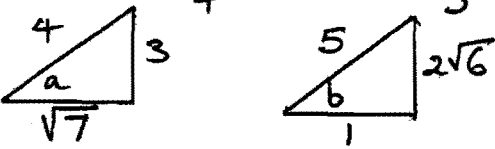
Solutions Yr 12 1/2 Yearly 2013

(ii) $2 \cos(x-30^\circ) = 1$
 $\cos(x-30^\circ) = \frac{1}{2}$ ①
 $x-30^\circ = 60^\circ, 300^\circ$
 $x = 90^\circ, 330^\circ$ ①

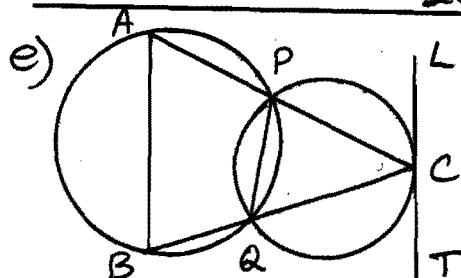
c) $\int \cos x \sin^4 x \, dx$

$\frac{d}{dx}(\sin^5 x) = 5 \cos x \sin^4 x$ ①

$\therefore \int \cos x \sin^4 x = \frac{1}{5} \sin^5 x + C$ ①

d) $\sin a = \frac{3}{4} \quad \cos b = \frac{1}{5}$
 ①

$\therefore \sin(a+b) = \sin a \cos b + \cos a \sin b$
 $= \frac{3}{4} \cdot \frac{1}{5} + \frac{\sqrt{7}}{4} \cdot \frac{2\sqrt{6}}{5}$
 $= \frac{3 - 2\sqrt{42}}{20}$ (or equivalent) ①



Ext L. Join PQ.

Let $\angle LCP = x^\circ$

$\therefore \angle LCP = \angle PQC = x^\circ$ ①
 (Alternate Segment Theorem)

$\angle PQC = \angle BAP = x^\circ$

(Exterior $\angle$ of a cyclic quad = interior opposite $\angle$). ①

$\therefore \angle BAP = \angle LCA = x^\circ$

(Since these alternate $\angle$'s are $= AB \parallel LT$) ①

12 a) ① $T = M + Ae^{kt}$
 $\Rightarrow Ae^{kt} = T - M$ ①

$\frac{dT}{dt} = k(T-M)$

$\frac{dT}{dt} = kAe^{kt}$ from ①
 $= k(T-M)$ ①

$\therefore T = M + Ae^{kt}$ is a soln of
 $\frac{dT}{dt} = k(T-M)$

(ii) $T = M + Ae^{kt} \quad M = 15$

when $t = 0 \quad T = 1000$

$\therefore 1000 = 15 + Ae^0$

$\therefore A = 985$

when $t=5$ $T=800$

$$\therefore 800 = 15 + 985 e^{5k}$$

$$e^{5k} = \frac{785}{985}$$

$$\therefore k = \frac{\ln\left(\frac{785}{985}\right)}{5}$$

$$k = -0.0453 \dots \textcircled{1}$$

find T when $t=15$

$$T = 15 + 985 e^{15(-0.0453 \dots)}$$

$$= 514^\circ \text{ (to the nearest deg.)}$$

(accept $514^\circ - 517^\circ$) $\textcircled{1}$
allowing for rounding.

b) Probability 4 of the players have number 1 is

$$\frac{1}{28} \frac{1}{28} \frac{1}{28} \frac{1}{28} = \frac{1}{28^4}$$

Nos $1 \rightarrow 8$

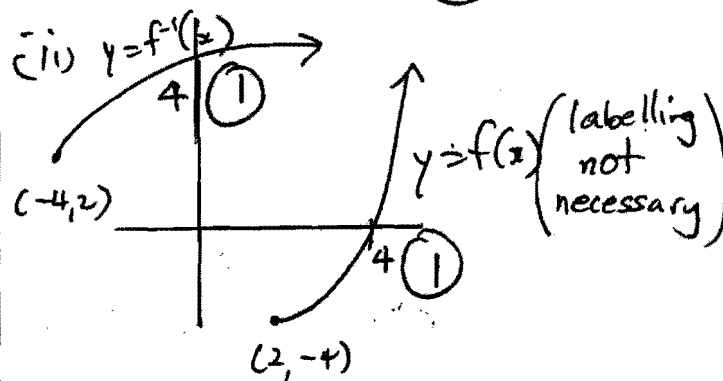
$$= 8 \times \frac{1}{28^4}$$

$$= \frac{8}{3024}$$

$$\therefore \text{Prob'g} = \frac{3024}{32^6} \rightarrow \textcircled{1}$$

c) (i) $y' = 0 \quad y = \frac{x^2 - 4x}{2}$
 $x > 2$

$\therefore$ largest domain containing $x=4$ is $x \geq 2$ $\textcircled{1}$



(iii) $D: f^{-1}(x): x \geq -4$ $\textcircled{1}$

(iv) inverse $x = y^2 - 4y$ $\textcircled{1}$

$$\therefore x + 4 = y^2 - 4y + 4$$

$$(y-2)^2 = x+4 \quad \textcircled{1}$$

$$y-2 = \pm \sqrt{x+4}$$

$$y = 2 \pm \sqrt{x+4}$$

but since $y \geq 2$ (because of restriction)

$$y = 2 + \sqrt{x+4} \quad \textcircled{1}$$

(v) Intersect on $y=x$

$$y=x \quad y=x^2-4x$$

$$x = x^2 - 4x$$

$$x^2 - 5x = 0$$

$$x(x-5) = 0 \quad x = 0, 5$$

but $x \geq 2 \quad \therefore x = 5$ $\textcircled{1}$ award mark at $x=5$
point is $(5, 5)$

$$f^{-1}(f(1)) = f^{-1}(f(3)) \text{ (because of symmetry + } x=1 \text{ is not in the restricted domain)}$$

$$= 3 \quad \textcircled{1} \text{ no need for reason}$$

Question 13. (See end of question for marking scale)

a) Prove $7^n + 3^n$ is $\div$ by 10 for odd integers n .

Step 1. Prove true for $n=1$

$$7^1 + 3^1 = 10 \text{ which is } \div \text{ by } 10$$

Step 2. Assume true for $n=k$ (k is odd)

$$\frac{7^k + 3^k}{10} = M \text{ (where } M \text{ is a pos. integer)}$$

$$7^k = 10M - 3^k \quad \textcircled{1}$$

Step 3. Prove true for $n=k+2$

$$\text{Show } \rightarrow 7^{k+2} + 3^{k+2} \text{ is } \div \text{ by } 10$$

$$7^2 \cdot 7^k + 3^2 \cdot 3^k$$

$$\text{from } \textcircled{1} = 49(10M - 3^k) + 9(3^k)$$

$$= 490M - 49(3^k) + 9(3^k)$$

$$= 490M - 40(3^k)$$

$$= 10(49M - 4(3^k))$$

Since 10 is a factor and m is a positive integer then $7^{k+2} + 3^{k+2}$ is $\div$ by 10.

Step 4 Proved true for $n=1$ and assumed true for $n=k+2$ prove true for $n=k+2 \therefore$ true for $n=1, n=3, n=5, \dots$ and for all odd integers n .

Marks - 4 Key components

1. Prove result for $n=1$
2. Clearly stating the assumption what is to be proven - (Must state m is a positive integer)
3. Using the assumption successfully in the proof.
4. Correctly proving the required statement

3 marks - successfully does all 4 components

2 marks - successfully does 3 of the 4 components but must include step 3.

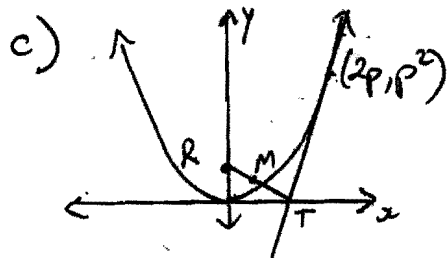
1 mark - does 2 of the 4 components successfully

$$b) x^3 - 5x^2 + 4x - 2 = 0$$

$$i) x + p + r = 5 \quad (1)$$

$$ii) x + p + r + p + r = 4 \quad (1)$$

$$iii) x^2 + p^2 + r^2 = (x + p + r)^2 - 2(xp + xr + pr) \\ = 5^2 - 2(4) \\ = 17 \quad (1)$$



$$x^2 = 4ay \\ x = 2at \quad y = at^2 \\ i) x = 2p \quad y = p^2 \quad \text{or} \quad x = 2p \quad y = p^2 \\ p = \frac{x}{2} \therefore y = \left(\frac{x}{2}\right)^2 \Rightarrow a = 1 \\ \therefore x^2 = 4y \quad (1) \therefore x^2 = 4(1)y \\ x^2 = 4y.$$

$$ii) y = \frac{x^2}{4} \quad (1) \\ y' = \frac{2x}{4} \quad \text{at } x = 2p \quad y' = \frac{4p}{4} = p$$

$$\text{Eq'n of tangent } y - p^2 = p(x - 2p)$$

$$y - p^2 = px - 2p^2 \\ y = px - p^2 \quad (1)$$

$$iii) \text{ at } T \quad y = 0 \therefore 0 = px - p^2 \\ x = \frac{p^2}{p} \rightarrow x = p \quad (1)$$

$$i) T(p, 0)$$

$$\text{RT has eq'n } y - 0 = \frac{1}{p}(x - p)$$

$$y = -\frac{x}{p} + 1 \quad (1)$$

$$\text{at } x = 0 \quad y = 1 \quad R(0, 1)$$

$$\text{now } x^2 = 4y \Rightarrow a = 1 \quad (1)$$

$$\therefore \text{ focus is } (0, 1)$$

$$\therefore R \text{ is the focus.}$$

$$(II) \text{ Midpoint}$$

$$M = \left(\frac{0+p}{2}, \frac{1+0}{2}\right)$$

$$= \left(\frac{p}{2}, \frac{1}{2}\right) \quad (\text{not essential})$$

$$\therefore \text{ locus } y = \frac{1}{2} \quad (1) \quad p \neq 0$$

$$III \quad \begin{cases} RT = \sqrt{p^2 + 1} \\ PT = \sqrt{(2p-p)^2 + (p^2-0)^2} \\ = \sqrt{p^2 + p^4} \\ = \sqrt{p^2(1+p^2)} \\ = p\sqrt{1+p^2} \end{cases} \quad (1)$$

$$\text{Area } \Delta = \frac{1}{2} \cdot p\sqrt{1+p^2} \cdot \sqrt{1+p^2} \\ \text{or equivalent } \leftarrow \frac{p(1+p^2)}{2} \quad (1)$$

14a) i)

Prove $\tan \theta = \sqrt{\frac{1 - \cos 2\theta}{1 + \cos 2\theta}}$

$$\text{RHS} = \sqrt{\frac{1 - (1 - 2\sin^2 \theta)}{1 + (2\cos^2 \theta - 1)}} \quad (1)$$

$$= \sqrt{\frac{2\sin^2 \theta}{2\cos^2 \theta}} \quad (1)$$

$$= \sqrt{\tan^2 \theta}$$

$$= \tan \theta$$

(ii) $\therefore \tan \frac{\pi}{8} = \sqrt{\frac{1 - \cos \frac{\pi}{4}}{1 + \cos \frac{\pi}{4}}}$

$$= \sqrt{\frac{1 - \frac{1}{\sqrt{2}}}{1 + \frac{1}{\sqrt{2}}}} \quad (1)$$

$$= \sqrt{\frac{\frac{\sqrt{2}-1}{\sqrt{2}} \times \frac{\sqrt{2}}{\sqrt{2}+1}}{1}} \quad (1)$$

$$= \sqrt{\frac{\frac{\sqrt{2}-1}{\sqrt{2}+1} \times \frac{\sqrt{2}-1}{\sqrt{2}-1}}{1}}$$

$$= \sqrt{\frac{(\sqrt{2}-1)^2}{2-1}}$$

$$= \sqrt{2}-1 \quad (1)$$

if they have
= $\sqrt{3-2\sqrt{2}}$
3 marks

b) $f(x) = x^3 + 2x - 4$

$$f'(x) = 3x^2 + 2$$

now $x^2 \geq 0$ for all $x \therefore 3x^2 + 2 \geq 0$ (1)

for all x

ie $f'(x)$ is increasing for all x .

hence the graph can only cut the

x axis at 1 place and hence (1)

there is only 1 root

c) (i) In $\Delta ACT \Rightarrow \tan 60^\circ = \frac{h}{AC}$

$$\therefore AC = \frac{h}{\tan 60^\circ} \Rightarrow AC = \frac{h}{\sqrt{3}}$$

In ΔCTB similarly $BC = \frac{h}{\tan 30^\circ}$

$$BC = h\sqrt{3}$$

(ii) $\angle CAB = 120^\circ$ let $AB = x$

$$\therefore BC^2 = AC^2 + AB^2 - 2AC \cdot AB \cos 120^\circ$$

$$3h^2 = \frac{h^2}{3} + x^2 - 2 \cdot \frac{h}{\sqrt{3}} \cdot x \cdot \left(-\frac{1}{2}\right) \quad (1)$$

$$3h^2 = \frac{h^2}{3} + x^2 + \frac{hx}{\sqrt{3}}$$

$$\therefore x^2 + \frac{hx}{\sqrt{3}} - \frac{8h^2}{3} = 0$$

$$\therefore x = \frac{-\frac{h}{\sqrt{3}} \pm \sqrt{\frac{h^2}{3} + 4 \cdot 1 \cdot \frac{8h^2}{3}}}{2}$$

$$= \left(\frac{-\frac{h}{\sqrt{3}} \pm \sqrt{\frac{33h^2}{3}}} \right) \div 2$$

$$x = \frac{-\frac{h}{\sqrt{3}} \pm h\sqrt{11}}{2} \text{ but } x > 0$$

$$\therefore x = \frac{-\frac{h}{\sqrt{3}} + \frac{h\sqrt{11}}{2}}{2}$$

$$= h \left(\frac{\sqrt{11}}{2} - \frac{1}{2\sqrt{3}} \right)$$

$$= h \left(\frac{\sqrt{33}-1}{2\sqrt{3}} \right)$$

$$\therefore \text{Ratio is } \frac{\sqrt{33}-1}{2\sqrt{3}} : 1 \quad (1)$$

(accept 1.37 : 1)

d) Method 1

$$\log_b xy^3 = m \quad \log_b x^3 y^2 = p$$

$$\therefore \log_b x + 3\log_b y = m \text{ and } 3\log_b x + 2\log_b y = p \quad (1)$$

$$3\log_b x + 2\log_b y = p$$

let $\log_b x = c$ and $\log_b y = d$.

$$\therefore c + 3d = m \quad (1)$$

$$3c + 2d = p \quad (2)$$

$$① \times 3 \quad 3c + 9d = 3m \quad (3)$$

$$③ - ② \quad 7d = 3m - p$$

$$\therefore d = \frac{3m - p}{7} \text{ sub in } ②$$

$$3c + 2 \left(\frac{3m - p}{7} \right) = p$$

$$\therefore c = \frac{9p - 6m}{21} = \frac{3p - 2m}{7}$$

$$\therefore \log_b y = \frac{3m-p}{7} \text{ and } \log_b x = \frac{3p-2m}{7} \quad (1)$$

Require $\log_b \sqrt{xy}$

$$= \frac{1}{2} (\log_b x + \log_b y)$$

$$= \frac{1}{2} \left(\frac{3p-2m}{7} + \frac{3m-p}{7} \right) \quad (1)$$

$$= \frac{1}{2} \left(\frac{2p+m}{7} \right)$$

$$\therefore \log_b \sqrt{xy} = \frac{2p+m}{14} \quad (1)$$

Method 2.

$$\log_b (xy^3) = m$$

$$\therefore b^m = xy^3$$

$$x = \frac{b^m}{y^3} \quad (1)$$

$$\log_b x^3 y^2 = p$$

$$\therefore b^p = x^3 y^2 \quad (2)$$

sub (2) into (1)

$$b^p = \left(\frac{b^m}{y^3} \right)^3 \cdot y^2$$

$$b^p = \frac{b^{3m}}{y^7}$$

$$\therefore y^7 = \frac{b^{3m}}{b^p} \Rightarrow y^7 = b^{3m-p}$$

$$y = \sqrt[7]{b^{3m-p}}$$

$$y = b^{\frac{3m}{7} - \frac{p}{7}} \quad (1)$$

sub into (1)

$$x = \frac{b^m}{\left(b^{\frac{3m}{7} - \frac{p}{7}} \right)^3}$$

$$= \frac{b^m}{b^{\frac{9m}{7} - \frac{3p}{7}}}$$

$$= b^{m - \frac{9m}{7} + \frac{3p}{7}}$$

$$x = b^{\frac{3p}{7} - \frac{2m}{7}} \quad (1) \quad (4)$$

$$\therefore \sqrt{xy} = \sqrt{b^{\frac{3m}{7} - \frac{p}{7}} \cdot b^{\frac{3p}{7} - \frac{2m}{7}}}$$

$$= \sqrt{b^{\frac{m}{7} + \frac{2p}{7}}}$$

$$= \left(b^{\frac{m}{7} + \frac{2p}{7}} \right)^{\frac{1}{2}}$$

$$= b^{\frac{m}{14} + \frac{2p}{14}}$$

$$\therefore \log_b \sqrt{xy} = \frac{m+2p}{14} \quad (1)$$